

Aarya Mehta

aarya.m.mehta@gmail.com | 972-834-6821 | Frisco, TX | US Citizen

Objective: Seeking an internship position for summer 2026 using my engineering skills in any location.

Education:

Purdue University

August 2025 - May 2028

First Year Engineering (Pursuing Aerospace)

West Lafayette, IN

GPA: 3.94/4.00

Skills: Fusion 360, NX 2306, Ansys Workbench, MATLAB, Python, Java, C, GitHub, Microsoft Office Suite

Experience:

LOx Inducer Manufacturing Responsible Engineer

September 2025 - Present

Purdue Undergraduate Rocket Propulsion Lab (PURPL)

- Member of student team designing and building *Marlin*, a 5000 lbf turbopump rocket engine scheduled for a full system hot-fire in March 2026.
- Currently on the pumps subteam, where I am leading the LOx inducer hardware workstream.
- Completed materials trade study to recommend 316L Stainless Steel for the LOx inducer based on identified parameters.
- Verified design viability through structural analysis using first-principles hand calculations for safety margins and thermal expansion, supported by detailed FEA.
- Created toolpath using CAM and machined complex geometry of inducer on 5 axis UMC 500 CNC mill.

Researcher/Senior Mentor

October 2023 - March 2025

NASA High School Aerospace Scholars (HAS)

- Collaborated with peers and NASA consultants to design and present a mission proposal for expanding the Artemis Gateway, including CAD modeling, applying engineering design, coding, risk mitigation, and exploration planning.
- Selected as a Senior Mentor for NASA HAS, providing technical and leadership guidance to junior scholars while completing training in communication and project management.

Co-President

September 2021 - May 2025

Mu Alpha Theta (Math Honor Society)

- Led 6-member officer team, grew membership to 150+ students through strategic outreach, and organized academic programs including AMC preparation, math tutoring, and quiz bowls.

Projects:

Dark Matter Research | *Python*

July 2023 - May 2025

- Co-authored a paper: Estimations of Dark Matter Fractions for ETGs using the Broken-Power-Law Model and Machine Learning Techniques (DOI: 10.1093/mnras/staf811)
- Published in Oxford's Monthly Notices of the Royal Astronomical Society with ~700 impressions.

Model Rocket | *CAD, OpenRocket*

January 2024 - March 2024

- Designed model rocket, adjusting factors such as weight distribution, shape and placement of fins, and the engine class.
- Ran simulations in Open Rocket of the rocket's launch, using the predicted flight path to test and fine-tune the design, aiming to maximize altitude.
- Assembled model rocket and launched it to an altitude of 177 meters.

Interests: Tennis, Basketball, Climbing, Astrophotography